

Grading Rubric: C# WPF .NET Desktop Application - Enterprise Readiness Assessment

Domain Tested	Core C# / .NET Engineering Objectives	Primary Toolchain	Crucial KPI & Log Markers	Scoring & Deduction Logic	Student Score (%)
1. Database & Query Performance	1. Measure EF Core query execution time for critical data operations. 2. Verify scoped DbContext disposal, no long-lived context instances held in ViewModels. 3. Validate database connection stability under normal and repeated operations.	EF Core query logging, Stopwatch timing, Visual Studio Diagnostic Tools	• Average query execution time (ms) • DbContext disposal pattern confirmed per operation • Database connection success rate across all features	100%: All queries execute under 500ms; scoped using var context pattern confirmed throughout; no connection failures during normal use. -30%: Long-lived _context instance field found in any ViewModel. -20%: Unhandled database exceptions or connection timeout errors.	
2. Async Operations & UI Responsiveness	1. Verify the UI thread is never blocked, all database and file operations must use async/await. 2. Validate AsyncRelayCommand usage for all long-running operations. 3. Confirm UI remains responsive with visible feedback during heavy operations.	Visual Studio Diagnostic Tools, UI thread monitoring, manual interaction testing	• UI thread block count (target: zero) • Percentage of DB-hitting commands using AsyncRelayCommand • Presence of progress indicators on operations exceeding 2 seconds	100%: Zero UI freezes detected; all database operations are async; progress feedback shown for long-running tasks. -40%: Synchronous blocking calls found on UI thread (.Result, .Wait(), or synchronous CopyTo()). -20%: Long-running operations with no progress indicator or user feedback.	

3. Data Integrity & File Handling	1. Verify CRUD operations maintain data consistency across all modules. 2. Test file upload, storage, and retrieval reliability for all supported file types. 3. Validate backup and restore functionality, data must match original after a full cycle.	Manual CRUD testing, file operation logs, before/after database comparison	• CRUD success rate across all modules • File upload and retrieval success rate • Backup-restore data integrity match percentage	100%: All CRUD operations produce consistent results; all file types upload and retrieve correctly; restored data matches source 100%. -30%: Data inconsistency found after any CRUD operation. • -20%: File retrieval failure or data loss after backup-restore cycle.	
4. System Stability & Code Quality	1. Validate memory stability during an extended session, no memory leaks from unreleased resources. 2. Verify proper implementation of IDisposable and using patterns throughout the codebase. 3. Confirm error handling is present on all critical operations, no unhandled exceptions crash the application.	Visual Studio Diagnostic Tools (memory snapshots), code review, manual extended session testing	• Memory usage stability over a 30-minute session • IDisposable and using pattern coverage • Unhandled exception count during full walkthrough (target: zero)	100%: Flat or stable memory curve over session; proper resource disposal confirmed; zero unhandled exceptions during full feature walkthrough. -40%: Memory growth exceeding 20% over session indicating a resource or context leak. -20%: Unhandled exceptions found during any standard operation.	
FINAL SYSTEM GRADE	Validate the C# WPF .NET desktop application's enterprise-level production readiness across all 4 domains and 12 parameters.			Average of all 4 Domain scores	